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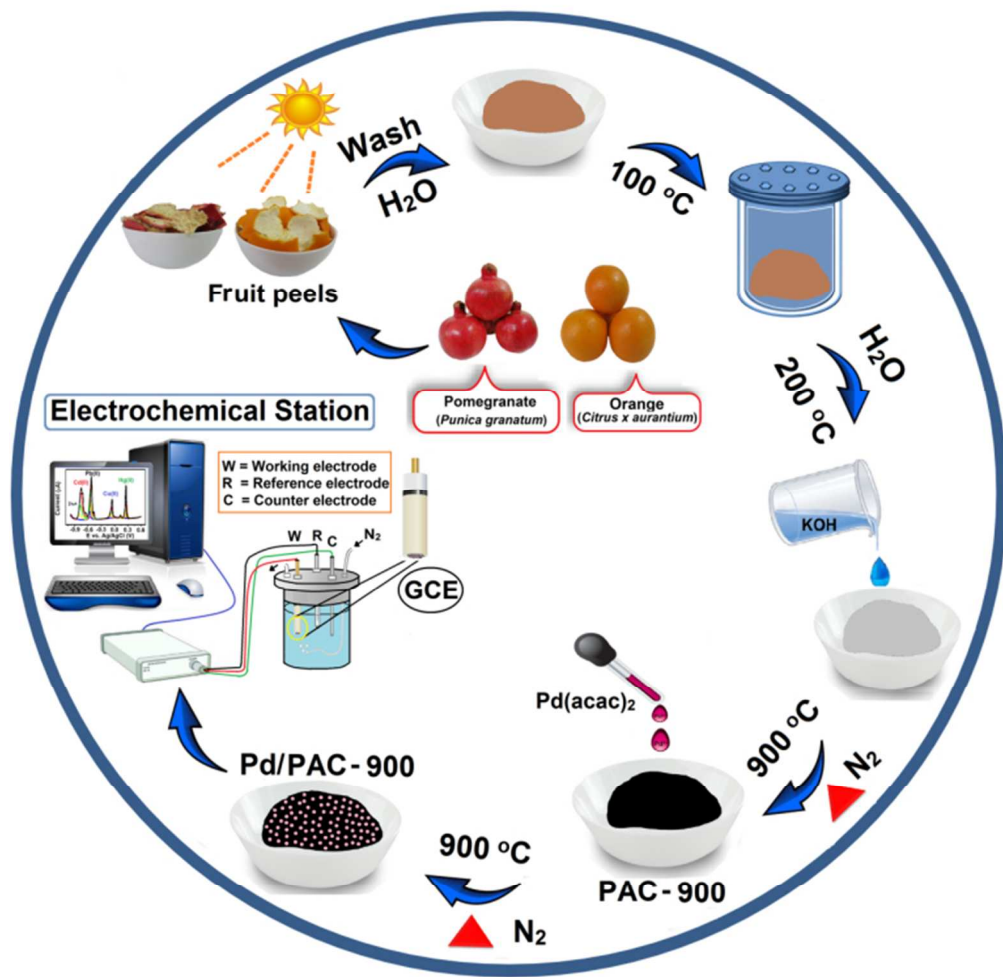
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ACS Appl. Mater. Interfaces, **Just Accepted Manuscript** • DOI: 10.1021/acsami.5b10050 • Publication Date (Web): 23 Dec 2015Downloaded from <http://pubs.acs.org> on December 29, 2015**Just Accepted**

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Electrochemical detection of Cd²⁺, Pb²⁺, Cu²⁺, and Hg²⁺ ions by Pd nanoparticles incorporated porous activated carbons (Pd/PACs)
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Palladium Nanoparticle Incorporated Porous Activated Carbon: Electrochemical Detection of Toxic Metal Ions

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ABSTRACT: A facile method has been developed for fabricating selective and sensitive electrochemical sensor for the detection of toxic metal ions, which invokes incorporation of palladium nanoparticles (Pd NPs) on porous activated carbons (PACs). The PACs, which were derived from waste biomass feedstock (fruit peels), possess desirable textural properties and porosities favorable for dispersion of Pd NPs (ca. 3–4 nm) on the graphitic PAC substrate. The Pd/PAC composite materials so fabricated were characterized by a variety of different techniques, such as X-ray diffraction, field-emission transmission electron microscopy, gas physisorption/chemisorption, thermogravimetric analysis, and Raman, Fourier-transform infrared, and X-ray photon spectroscopies. The Pd/PAC-modified glassy carbon electrodes (GCEs) were exploited as electrochemical sensors for the detection of toxic heavy metal ions, viz. Cd^{2+} , Pb^{2+} , Cu^{2+} , and Hg^{2+} , which showed superior performances for both individual as well as simultaneous detections. For simultaneous detection of Cd^{2+} , Pb^{2+} , Cu^{2+} , and Hg^{2+} , a linear response in ion concentration range of 0.5–5.5, 0.5–8.9, 0.5–5.0, and 0.24–7.5 μM , with sensitivity of 66.7, 53.8, 41.1, and 50.3 $\mu\text{A } \mu\text{M}^{-1} \text{ cm}^{-2}$, and detection limit of 41, 50, 66 and 54 nM, respectively, were observed. Moreover, the Pd/PAC-modified GCEs is also show perspective applications in detection of metal ions in real sample, as illustrated in this study for a milk sample.

KEYWORDS: *palladium nanoparticles, fruit peels, porous activated carbon, toxic metal ions, cyclic voltammetry, sensors*

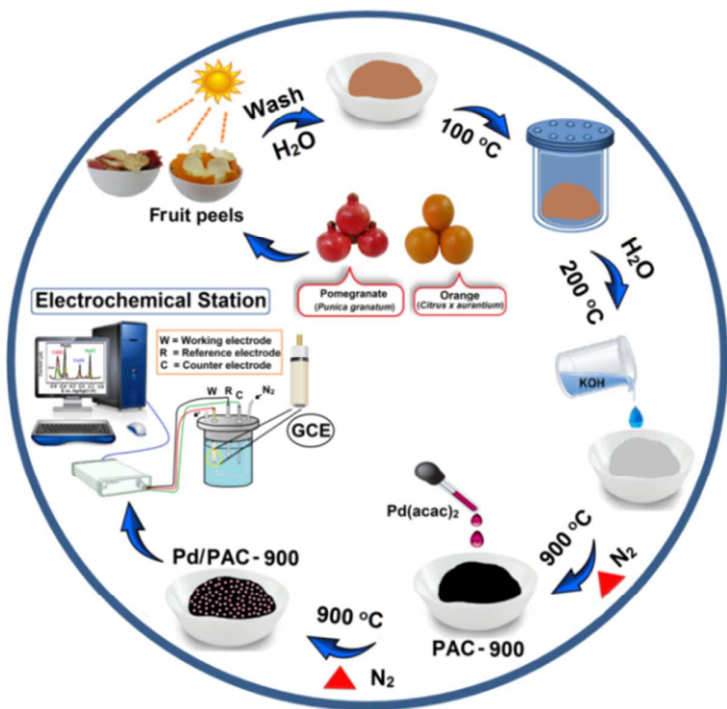
1. INTRODUCTION

Porous activated carbons (PACs) derived from biomass feedstock have received considerable attentions recently owing to their growing R&D interests in areas related to environment, energy, and sensing,^{1–4} especially for practical applications in catalysis, gas adsorption/separation, fuel and energy storages, fuel cells, dye sensitized solar cells (DSSCs), electrochemical sensors, and chemical/biological sensing of toxic metal ions.^{5–13} Several new methodologies have been developed to synthesize PACs from cheap and renewable sources such as industrial and agricultural wastes^{14–18} utilizing various activation procedures aiming to achieve substrates with higher porosities and surface areas.^{19–22} For example, PACs are commonly prepared through physical methods by pyrolyzing the carbon precursors under inert atmosphere to remove most of non-carbon elements, followed by an activation procedure in the presence of a suitable oxidizing agent (e.g., O₂, CO₂, or steam) to render carbonization at elevated temperatures (typically, 600–1200 °C). Likewise, PACs may also be fabricated *via* a chemical method, during which the biowaste feedstock is first mixed with an activating agent (e.g., KOH, K₂CO₃, Na₂CO₃, H₃PO₄, ZnCl₂, FeCl₃, and so on), followed by the carbonization procedure at 400–900 °C under inert atmosphere. In this context, chemical activation process is superior over the physical activation method mainly due to the lower graphitization temperature required as well as the formation of PACs with high porosities (consisting of both micro- and mesopores), surface area (typically exceeding 1500 m² g^{−1}, and large pore volumes (≥ 0.8 cm³ g^{−1}), which are particularly advantageous for applications as electrochemical sensor for the detection of biomolecules or toxic metal ions.^{23–28} For examples, palladium nanoparticles (Pd NPs) decorated carbon-based composite modified glassy carbon electrodes (GCEs) have been exploited as potential electrochemical sensors for the detection of gases (such as hydrogen²⁹ and methane³⁰) and biomolecules such as uric acid,³¹ glucose,³² dopamine,^{26,31,33} ascorbic acid,^{31,34} and hydrazine.³⁵

Heavy metal ions such as arsenic (As³⁺), chromium (Cr³⁺ or Cr⁶⁺), cobalt (Co²⁺), nickel (Ni²⁺), copper (Cu²⁺), zinc (Zn²⁺), cadmium (Cd²⁺), lead (Pb²⁺), silver (Ag⁺), and mercury (Hg²⁺), and which may accumulate in the body through food chains, are highly toxic and hazardous pollutants to animals and human, leading to adverse effects on immune, central nervous, and reproductive systems even at a trace concentration.^{36–39} Hence, the R&D on detection of heavy metal ions focus

predominantly on relevant remediation and detection technologies. As such, a wide variety of detection methods have been made available for toxins detection, such as inductively coupled plasma-mass spectrometry (ICP-MS), atomic absorption spectroscopy (AAS), X-ray fluorescence (XRF) spectrometry, atomic emission spectrometry (AES) and so on.⁴⁰⁻⁴³ The main drawbacks of these optical techniques lie in expensive apparatus and time-consuming operating procedures. Moreover, owing to their coexisting nature, the simultaneous detection of heavy metal ions while rendering high sensitivity and selectivity remains a challenging task. In this regard, electrochemical detection has been proven to be a powerful, quick, facile, and cost-effective method.^{6,44-48} Meanwhile, scientists focusing on relevant research fields are invoking more environmental benign techniques to fabricate metal nanoparticles (NPs) supported PACs composites as modified GCEs for electrochemical sensors.

We report herein a facile, template-based procedure for the fabrication of Pd NPs embedded PAC nanocomposites for application as electrochemical sensor for efficient detection of toxic metal ions, viz. Cd^{2+} , Pb^{2+} , Hg^{2+} , and Cu^{2+} , which are stable and highly toxic. Thus, it is highly desirable to develop a sensitive method for the detection of these toxic metal ions.^{6,7,39,49-53} As illustrated in Scheme 1, the Pd/PAC composites so fabricated exhibit excellent sensing properties for sensitive and selective detections of metal ions, even in real samples. This is ascribed due to the uniformly dispersed Pd NPs embedded on the matrix of PACs, which were derived from biowaste materials, namely pomegranate and orange fruit peels.



Scheme 1. Schematic Illustration of Applications of Pd/PAC Materials for Detection of Toxic Heavy Metals

2. EXPERIMENTAL SECTION

2.1. Materials and Chemicals. Pomegranate (*Punica granatum*) and orange (*Citrus x aurantium*) fruit peels were collected from a local market in New Taipei City, Taiwan. Palladium(II) acetylacetonate ($\text{Pd}(\text{acac})_2$, 96%), potassium hydroxide (KOH), cadmium(II) chloride (CdCl_2 , 98%), lead(II) chloride (PbCl_2 , 98%), copper(II) chloride (CuCl_2 , 98%), and mercury(II) chloride (HgCl_2 , 98%) were obtained commercially (Sigma-Aldrich) and used without further treatment. All other chemicals used were analytical grades, and all solutions were prepared by using ultrapure water (Millipore).

2.2. Synthesis of Pd/PAC Nanocomposite. For the synthesis of PAC, the pomegranate and orange fruit peels) were first thoroughly washed with deionized water to remove the undesirable impurities, then, sun dried for 2 d. Equal amount of the dried biowastes (0.25 kg each) was mixed, ground, and sieved to obtained fine powders with an average particle size of ca. 1–2 mm, then heat treated at 100 °C for 10 h in an autoclave to remove water soluble phenolic and other organic compounds. Subsequently, the resultant powders were mixed with KOH pellets in deionized water and heated at 200 °C for 24 h in an autoclave, then, vacuum dried. The greyish

powdered material was further subjected to carbonization treatment at different temperatures ($T = 700, 800, 900$, and $1000\text{ }^{\circ}\text{C}$) for 3 h under flowing N_2 , the final products so obtained are denoted as PAC-T.

The Pd/PAC nanocomposites were prepared following the procedures: (i) chemical activation of PACs by using KOH as the agent, then, further oxidized with 2 M nitric acid (HNO_3) for 3 h at $80\text{ }^{\circ}\text{C}$ under an ultrasonic bath; (ii) mixing PACs in ethanol (3.0 mL) solution of the metal precursor $\text{Pd}(\text{acac})_2$ under sonication for 1 h; (iii) carbonization treatment at $900\text{ }^{\circ}\text{C}$ under N_2 atmosphere. The final Pd/PAC products loaded with 1.0 and 1.5 wt% Pd are labelled as $\text{Pd}_{1.0}/\text{PAC-900}$ and $\text{Pd}_{1.5}/\text{PAC-900}$, respectively.

To fabricate the Pd/PAC-modified electrodes, the Pd/PAC nanocomposite was dispersed in doubly distilled water under sonication for 1 h. Subsequently, ca. 6 μL of the prepared suspension was drop casted onto the surface of a polished glassy carbon electrode (GCE), followed by drying at $60\text{ }^{\circ}\text{C}$ for 0.5 h.

2.3. Characterization Methods. All powdered X-ray diffraction (XRD) experiments were recorded on a PANalytical (X'Pert PRO) diffractometer using $\text{Cu K}\alpha$ radiation ($\lambda = 0.1541\text{ nm}$). Nitrogen adsorption/desorption isotherm measurements were carried out with a Quantachrome (Autosorb-1) volumetric physisorption analyzer at $-196\text{ }^{\circ}\text{C}$ (77 K). Prior to each measurement, the sample was purged with flowing N_2 at $150\text{ }^{\circ}\text{C}$ for at least 12 h to remove pre-adsorbed water. Pore size distributions of various samples were derived from the adsorption branches of isotherms using the non-local density functional theory (NLDFT) method. The morphologies of various samples were studied by field-emission transmission electron microscopy (FE-TEM) at room temperature ($25\text{ }^{\circ}\text{C}$) using an electron microscope (JEOL JEM-2100F) operating at an acceleration voltage of 200 kV. Elemental compositions of different samples were determined by an energy-dispersive X-ray (EDX) analyzer equipped along with the FE-TEM apparatus. X-ray photoelectron spectroscopy (XPS) measurements were performed using an ULVAC-PHI PHI 5000 Versa Prob apparatus. Thermogravimetric analyses (TGA) were conducted on a Netzsch TG-209 instrument under air atmosphere. Ultraviolet-visible (UV-vis) absorption spectra were collected with a SPECORDS100 diode-array spectrophotometer. Fourier-transform infrared (FT-IR) spectra were recorded using a Bruker IFS-28 spectrometer in the region of $4000\text{--}400\text{ cm}^{-1}$ with a spectral resolution of 2 cm^{-1} using the KBr pellet method at room temperature. All Raman spectra were

obtained on a Jobin Yvon T64000 spectrometer equipped with a charge coupled device (CCD) detector cooled with liquid nitrogen. The backscattering signal was collected with a microscope using an Argon ion laser with a centering wavelength of 488 nm. Cyclic voltammetry (CV) and differential pulse voltammetry (DPV) studies were performed on an electrochemical analyzer (CHI 900; CH instruments). The 0.1 M acetate buffer solution (ABS) was used as the supporting electrolyte (pH = 5.0), which was prepared by using acetic acid (HAc) and sodium acetate (NaAc). A conventional three-electrode cell system was utilized using a modified glassy carbon electrode (GCE) as the working electrode, Ag/AgCl (in saturated KCl) as the reference electrode, and a platinum wire as the counter electrode.

3. RESULTS AND DISCUSSION

3.1. Sample Characterization. The XRD spectra of as-prepared PAC-T (T = 700, 800, and 900 °C) samples and Pd/PAC-900 nanocomposites loaded with 1.0 and 1.5 wt% of Pd NPs are depicted in Figure 1a. All pristine PAC samples exhibited two broad peaks at $2\theta = 23.7$ and 43.2° , which may be indexed to the characteristic diffraction of the (002) and (100) plane, respectively. The former peak may be attributed to the hexagonal graphite structure with a d -spacing of 0.334 nm.¹² Upon incorporating Pd NPs onto the PAC substrate, four additional sharp peaks at $2\theta = 40.3, 46.8, 68.3,$ and 82.4° respectively corresponding to the (111), (200), (220), and (311) index planes of the face-centered cubic (fcc) crystalline Pd metal (JCPDS 05-0681) were observed.⁵⁴ Further analysis by the Scherrer equation⁵⁵ based on the (111) peak revealed an average crystalline size of 4–5 nm for the Pd NPs on both Pd_{1.0}/PAC-900 and Pd_{1.5}/PAC-900 composite samples. Moreover, the d -spacing (d_{002}), Lateral size (L_a), and stacking height (L_c) of the PAC crystallite may be derived from the XRD data, as depicted in Table S1 of the Supporting Information (hereafter denoted as SI). That a consistent decrease in lattice spacing d_{002} with increasing carbonization temperature was observed, which is accompanied by the simultaneous increases in both L_a and L_c , indicating a gradual increase in crystallinity of the PAC materials.⁵⁶ In other word, a consistent increase in degree of stacking order towards a more graphitic structure with increasing graphitization temperature may be inferred.

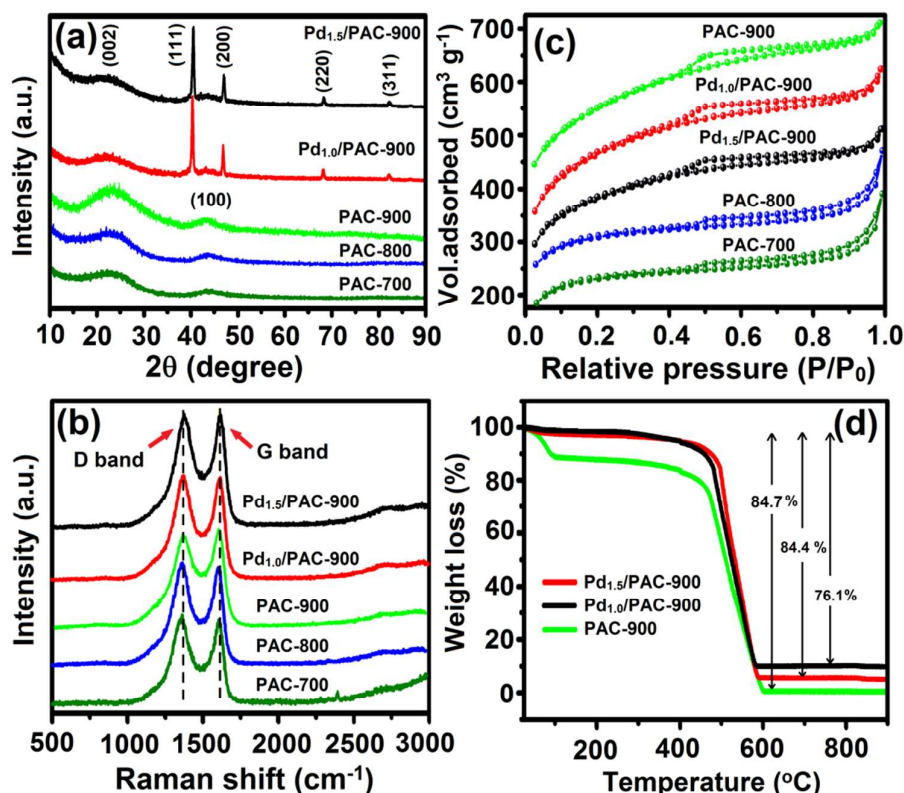


Figure 1. (a) XRD patterns, (b) Raman spectra, (c) N₂ adsorption/desorption isotherms, and (d) TGA curves of the as-prepared PACs, and Pd/PAC-900 nanocomposites samples.

The above results obtained from XRD are consistent with that deduced from Raman spectroscopy. The spectra observed for various PAC and Pd/PAC-900 samples revealed the presence of two prominent features at 1368 and 1603 cm⁻¹ (Figure 1b), which may be ascribed due to vibration bands of carbons in disordered graphite (D band) and the E_{2g} mode of the graphite (G band) relating to vibration of sp² bonded carbons in two-dimensional (2D) hexagonal lattice.⁵⁷ The D band is normally correlated with disordered turbostratic/defective and non-graphitic carbons, whereas the G band is associated with intermolecular shear vibrations between graphitic carbon layers.⁵⁸ Accordingly, the G to D band intensity ratios (I_G/I_D), which may be used to assess the crystalline structure of the graphitic carbons, obtained from various samples are depicted in Table 1. The I_G/I_D ratio observed for the PACs increased slightly from 0.97 to 1.02 as the carbonization temperature increased from 700 to 900 °C, indicating a gradual increase in crystallinity, in good agreement with the XRD results (Table S1; SI). That negligible change in I_G/I_D ratio was observed for the Pd/PAC-900 (1.01) compared to the PAC-900 (1.02), indicating that the carbon support retained a highly graphitized structure even after the incorporation of Pd NPs.

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Figure 1c shows the N₂ adsorption/desorption isotherms of various samples, which all revealed the typical type IV isotherm (IUPAC classification). The presence of a hysteresis loop extending from $P/P_0 = 0.45$ to 0.9 together with the abrupt increase in adsorption in the Henry's law ($P/P_0 < 0.05$) region, indicating the coexistence of both micro- and mesoporosities in all samples. Further data analyses revealed a consistent increase in porosity of the as-prepared PACs with increasing carbonization temperature from 700 to 900 °C. This may be seen by the increases in both surface area (S_{BET}) and total pore volume (V_{tot}) with increasing temperature (Table 1). However, further increasing the activation temperature to 1000 °C resulted in a notable decreases in both S_{BET} and V_{tot} (Table 1), indicating a degradation in both porosity and structural integrity of the graphitic carbon (Figure S1; SI).

The pore size distributions (PSDs) derived from the NLDFT method are depicted in (Figure S2a; SI). It is clear that with the exception of the PAC-700, which showed a weak PSD profile, the other PACs and Pd/PAC-900 samples revealed distribution of micro- and mesopore sizes ranging from 0.5 to 4.2 nm. An average pore size (d_{DFT} ; see Table 1) of ca. 3.9 nm may be inferred for the PAC-900. Upon incorporating Pd NPs onto the surfaces of PAC, notable decreases in textural parameters were observed.^{12,59} For examples, the value of S_{BET} , V_{tot} , and d_{DFT} observed for the PAC-900 sample (i.e., 1627 m² g⁻¹, 1.69 cm³ g⁻¹, and 3.9 nm, respectively) was found to decrease respectively to 1487 m² g⁻¹, 0.95 cm³ g⁻¹, and 3.4 nm for the Pd_{1.5}/PAC-900 (Table 1).

Table 1. Physical Properties of the As-prepared PAC and Pd/PAC Composite Materials

sample	Pd loading (wt%)	M_p (nm) ^a	S_{BET} (m ² g ⁻¹) ^b	S_{micro} (m ² g ⁻¹) ^c	V_{tot} (cm ³ g ⁻¹) ^b	V_{micro} (cm ³ g ⁻¹) ^c	V_{meso} (cm ³ g ⁻¹) ^d	d_{DFT} (nm) ^e	I_G/I_D ^f
PAC-700			963	465	0.84	0.33	0.51	2.8	0.97
PAC-800			1330	490	0.97	0.38	0.59	3.1	0.98
PAC-900			1627	558	1.69	0.45	1.24	3.9	1.02
PAC-1000			1432	309	1.73	0.34	1.39	4.2	1.00
Pd _{1.0} /PAC-900	1.0	3.5 ± 0.2	1589	506	1.08	0.36	0.72	3.5	1.01
Pd _{1.5} /PAC-900	1.5	3.8 ± 0.2	1487	494	0.95	0.28	0.67	3.4	1.01

^aAverage metal particle size determined by FE-TEM analysis. ^bBrunauer-Emmet-Teller surface area (S_{BET}) and total pore volume (V_{tot}) calculated at $P/P_0 = 0.99$. ^cMicroporous surface area and pore volume obtained from t -plot analysis. ^dMesoporous pore volume ($V_{\text{meso}} = V_{\text{tot}} - V_{\text{micro}}$). ^ePore size derived from the non-local density functional theory (NLDFT) method. ^fPeak intensity ratio of the G and D bands obtained from Raman spectrum.

The thermal properties of Pd/PAC composite samples were further examined by TGA; the results are displayed in Figure 1d and their corresponding differential thermal analysis (DTA) curves are shown in Figure S2b (SI). The first weight-loss observed for the PAC-900 and Pd/PAC-900 samples below 100 °C is ascribed due to desorption of physisorbed water. The abrupt weight-loss in the range of 450–550 °C may be attributed to the decomposition of the carbon framework, whereas that occurred above 600 °C is most likely due to decompositions of cellulosic and lignocellulosic fragments.^{1,60}

Figure 2 displays the FE-TEM images of the as-prepared PAC-900 and PD/PAC-900 materials at various magnifications. The images for the latter two samples clearly show the presence of Pd NPs, which are well-dispersed in the carbon matrix (Figure 2d–f) with an average particle size of ca. 3–4 nm (Table 1). Additional measurements by selected-area electron diffraction (SAED) confirm the crystalline fcc structure of the Pd NPs (insets, Figures 2e and 2f).^{12,61} Further EDX analyses also revealed the anticipated presences of C, O, and Pd elements in the Pd/PAC composite materials (Figure S2c; SI). The above results are in line with the XPS spectra obtained from the PAC-900 and Pd_{1.5}/PAC-900 samples (Figure 3), which revealed characteristic peaks at a binding energy of 533.1, 335.1 and 340.4, and 284.2 eV corresponding to O 1s, Pd 3d_{5/2}, Pd 3d_{3/2}, and C 1s spin-orbits, respectively.^{11,62}

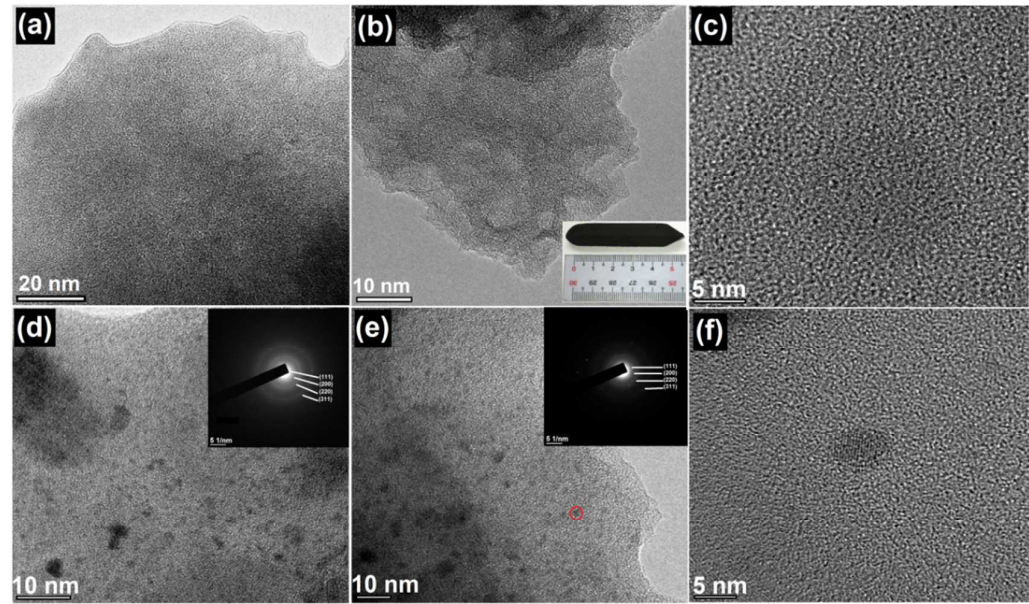


Figure 2. FE-TEM images of (a–c) PAC-900, (d) Pd_{1.0}/PAC-900, and (e,f) Pd_{1.5}/PAC-900 samples. Insets: corresponding (b) digital image and (d,e) SEAD patterns.

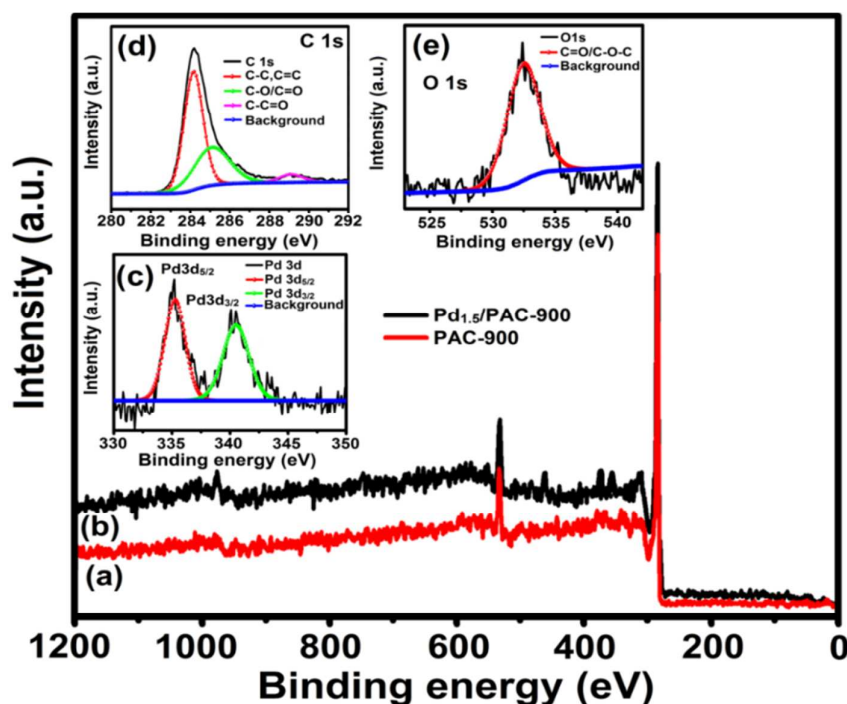


Figure 3. XPS spectra of (a) PAC-900 and (b) Pd_{1.5}/PAC-900, and expanded profiles in the (c) Pd 3d_{5/2} and Pd 3d_{3/2}, (d) C 1s, and (e) O 1s regions.

The pathway by which Pd NPs were incorporated onto the PAC matrix was further investigated by FT-IR and UV-vis spectroscopy. This is done by comparing the spectra recorded on PAC-900 before and after mixing with ethanol solution of the metal precursor, namely Pd(acac)₂. The pristine Pd(acac)₂ exhibited pronounced characteristic peaks at ca. 1430 and 1528 cm⁻¹, which may be attributed to stretching of aromatic C=C bond and carbonyl groups, respectively. As also shown in Figure S3a (SI), this latter band at 1528 cm⁻¹ diminished upon adsorbing Pd(acac)₂ onto the PAC-900, indicating the conversion of C=O groups into C-O groups, which was originally presented in the PAC-900 (1050 cm⁻¹). Upon introducing the metal precursor, a notable red shift in this C-O vibrational band was observed for the Pd(acac)₂/PAC-900 substrate, indicating its active involvement during incorporation of Pd NPs onto the carbon matrix.⁶³ Likewise, the three adsorption bands ($\lambda_{\text{max}} = 332$, 252, and 224 nm) observed in the UV-Vis spectrum of Pd(acac)₂ (Figure S3b; SI) decreased gradually with increasing sonication time after mixing with the PAC-900. This also indicates the strong interaction and good immersion of Pd(acac)₂ within the hydrophilic carbon matrix.⁶⁴

3.2. Simultaneous Detection of Cd(II), Pb(II), Cu(II), and Hg(II) Ions.

Figure 4a displays the CV profiles of the bare, Pd_{1.0}/PAC-900, and Pd_{1.5}/PAC-900 modified GCE in 0.1 M acetate buffer solution (ABS) containing mixture of Cd²⁺ (3.5 μM), Pb²⁺ (3.2 μM), Cu²⁺ (6.0 μM), and Hg²⁺ (3.2 μM) ions, respectively at 50 mV s⁻¹. The bare GCE exhibited only weak peak current responses accountable for Cd²⁺, Cu²⁺, and Hg²⁺ (*vide infra*), indicating its inferior performance for simultaneous detection of heavy metal ions. Upon progressive incorporation of Pd onto the PAC, the Pd/PAC-900 modified GCEs showed sharp and well-defined peaks at -0.69, -0.48, -0.21, and -0.32 V, which corresponded to the distinct responses for the Cd²⁺, Pb²⁺, Cu²⁺, and Hg²⁺ ions, respectively. The enhanced electrochemical performance observed for the Pd/PAC modified GCEs is clearly due to a synergistic effect between Pd NPs and the PAC-900. Moreover, Compared to its counterpart, the Pd_{1.5}/PAC-900 modified GCE showed the lowest charge transfer resistance, suggesting that the Pd NPs readily help to enhance the conduction pathway when anchored on surfaces of the PAC-900 substrate.

To explore the sensitivity and detection limit of the Pd/PAC modified GCE for simultaneous detection of Cd²⁺, Pb²⁺, Cu²⁺, and Hg²⁺ ions, we performed DPV measurements, which is well-known for its faster response compared to the other techniques, such as stripping voltammetry. As shown in Figure 4b, the DPV profiles observed for the Pd_{1.5}/PAC-900 modified GCE under varied metal ion concentrations (0.5–8.9 μM) showed four well-resolved response peaks at -0.79, -0.54, -0.07, and +0.27 V accountable for Cd²⁺, Pb²⁺, Cu²⁺, and Hg²⁺ ions, respectively. While the peak intensities clearly increase linearly with increasing ion concentration regardless of the type of the metal analyte (Figure 4c), some weak and shouldering peaks were also observed, indicating the formation of intermetallic compounds.⁶ For example, the well-resolved weak peak at ca. +0.15 V is most likely due to the formation of Cu–Hg intermetallic compound. Accordingly, the sensitivity and limit of detection (LOD) for the Pd_{1.5}/PAC-900 modified GCE for simultaneous detection of these four metal ion species may be derived, as summarized in Table 2.

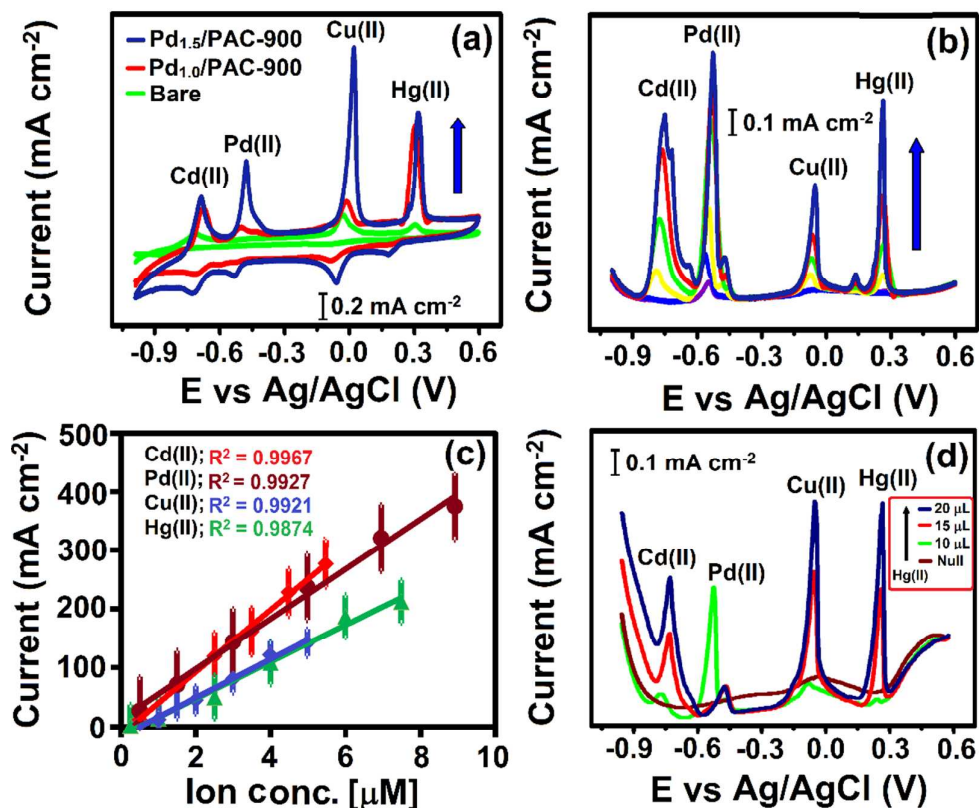


Figure 4. (a) CV profiles of bare, Pd_{1.0}/PAC-900, and Pd_{1.5}/PAC-900 modified GCEs in a mixture of Cd²⁺ (3.5 μM), Pb²⁺ (3.2 μM), Cu²⁺ (6.0 μM), and Hg²⁺ (3.2 μM) ions. (b) DPV curves observed for the Pd_{1.5}/PAC-900 modified GCE under different metal ion concentrations (0.5–8.9 μM) and (c) corresponding linear calibration plots of peak current vs metal ion concentration. All measurements were recorded in electrolyte: 0.1 M ABS (pH 5.0) at a scan rate of 50 mV s⁻¹. (d) Similar to (b) but in the presence of a mixture of a real milk sample with metal ion analytes: Cd²⁺ (2.4 μM), Pb²⁺ (2.9 μM), Cu²⁺ (2.9 μM), and varied amounts of Hg²⁺ (10–20 μL). Electrolytes (d): 5 mL of milk extract with 5 mL ABS (pH = 5) solution.

Table 2. Summary of Analytical Parameters for Simultaneous and Individual Detection of Various Metal Ions over the Pd_{1.5}/PAC-900 Modified GEC

performance	simultaneous detection				individual detection			
	Cd ²⁺	Pb ²⁺	Cu ²⁺	Hg ²⁺	Cd ²⁺	Pb ²⁺	Cu ²⁺	Hg ²⁺
linear range (μM)	0.5–5.5	0.5–8.9	0.5–5.0	0.24–7.5	0.5–12.8	0.5–22.4	0.5–11.8	0.5–5.0
sensitivity (μA μM ⁻¹ ·cm ⁻²)	66.7	53.8	41.1	50.3	72.9	109.1	21.8	103.3
detection limit (nM)	41	50	66	54	38	25	130	27

3.3. Individual Detection of Various Metal Ions. The capability and performance of the Pd_{1.5}/PAC-900 modified GCE for individual detection of a specific metal ion while under the presence and influence of other ions were also examined. Similar experimental parameters used for simultaneous detection were adopted except, in this case, while varying the concentration of the targeted metal ion, the concentrations of the other metal ions were kept fixed at 2 μ M. For example, the DPV profile observed for detection of Pd²⁺ shown in Figure 5a were recorded under varied concentrations of Cd²⁺ (0.5–12.8 μ M) while in presence of a fixed amount (2 μ M) of the other metal ions, namely Pb²⁺, Cu²⁺, and Hg²⁺. Upon increasing the concentration of Cd²⁺, the slight decrease in the observed peak current of Pb²⁺ is due to formation of intermetallic compound, which verify the detection sensitivity of the Pd_{1.5}/PAC-900 modified GCE. Moreover, a linear dependence between the peak current intensity with ion concentration was observed (inset, Figure 5a). Likewise, similar conclusions may be inferred for selective detections of Pb²⁺, Cu²⁺, and Hg²⁺, as shown in Figures 5b–d, respectively. Accordingly, the calculated sensitivities and detection limits observed for selective detection of Cd²⁺, Pb²⁺, Cu²⁺, and Hg²⁺ ions are tabulated and compared with those obtained from simultaneous detection, as shown in Table 2. It is indicative that comparable analytical parameters were obtained for selective and simultaneous detection, indicating that the Pd_{1.5}/PAC-900 modified GCE is suitable for practical applications.

3.4. Real Sample Tests. To further illustrate the feasibility of Pd/PAC composite for real sample applications, the Pd_{1.5}/PAC-900 modified GCE is further employed for the detection of toxic metal ions in a real milk sample, which normally contains traceable amounts of Cd²⁺ and Pb²⁺ ions by itself.^{65,66} As shown in Figure 4d, the DPV profile of the pure milk sample itself show practically no metal ion signals. However, upon introducing a mixture of various metal ions (2.4 μ M Cd²⁺, 2.9 μ M Pb²⁺, 2.9 μ M Cu²⁺) into the real milk sample (5 mL) while varying the concentration of Hg²⁺ (10–20 μ L), oxidation peaks corresponding to various ion analytes were observed. In addition, the oxidation peak currents associated with various metal ions were found to increase with increasing Hg²⁺ concentration, indicating the presence of intermetallic interactions. The above results clearly demonstrate that the Pd/PAC modified GCE is suitable for detection of heavy metal ions even in the presence of a real sample.

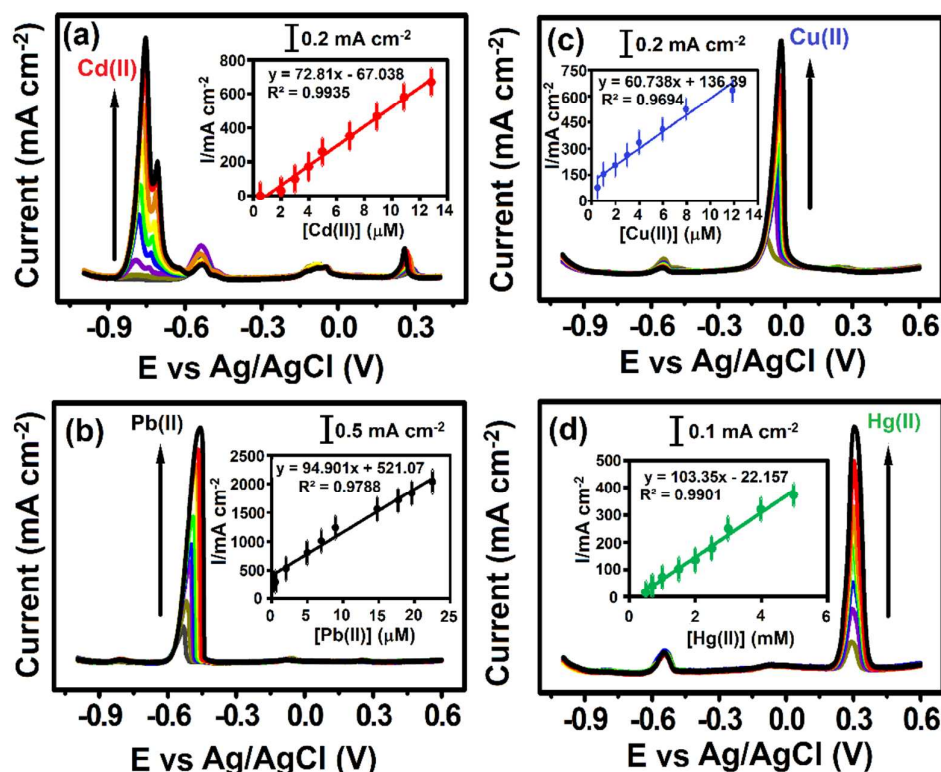


Figure 5. DPV response for the $\text{Pd}_{1.5}/\text{PAC-900}$ modified electrodes for selective detection of metal ions at different concentrations in μM : (a) Cd^{2+} (0.5–12.8), (b) Pb^{2+} (0.5–22.4), (c) Cu^{2+} (0.5–11.8), and (d) Hg^{2+} (0.5–5.0). Electrolyte: 0.1 M ABS (pH 5.0) at a scan rate of 50 mV s^{-1} . Insets: corresponding linear calibration plots of peak current vs ion concentration.

4. CONCLUSIONS

An easy, cost-effective, and eco-friendly synthesis route for the preparation of Pd NPs incorporated porous activated carbon (Pd/PAC) materials has been developed and successfully employed for electrochemical detection of heavy metal ions, namely Cd^{2+} , Pb^{2+} , Cu^{2+} , and Hg^{2+} . Moreover, the $\text{Pd}_{1.5}/\text{PAC-900}$ -modified GCE, which contains 1.5 wt% Pd and was activated at 900°C , was found to exhibit excellent catalytic activity, selectivity, sensitivity, and low detection limit for selective as well as simultaneous detections of heavy metal ions, even in a real milk sample. The facile synthesis route developed herein for the preparation of metal NPs incorporated PACs from renewable biomass wastes hence is not only cost-effective and environmental friendly but also feasible for large-scale productions. Moreover, the Pd/PAC-modified electrodes, which exhibit excellent performances toward selective and sensitive detection of heavy metal ions surpassing most other modified GCEs (Table 3), also show perspective and practical industrial applications.

Table 3. Performances of Various Modified Electrodes for Simultaneous Detection of Heavy Metal Ions*

modified electrode	method	metal ion (linear range)	detection limit	ref
SNAC/GCE	DPV	Cd ²⁺ (0.09–4.8); Pb ²⁺ (0.09–5.7); Cu ²⁺ (0.09–4.8); Hg ²⁺ (0.09–0.99) μM	Cd ²⁺ (24.4); Pb ²⁺ (5.7); Cu ²⁺ (23.2); Hg ²⁺ (24.6) nM	6
HAP/Ag NP-ZSM-5	SWSV	Cd ²⁺ (0.5–1600); Pb ²⁺ (0.6–1600); As ³⁺ (0.9–1800); Hg ²⁺ (0.8–1800) ppb	Cd ²⁺ (0.1); Pb ²⁺ (0.1); As ³⁺ (0.2); Hg ²⁺ (0.2) ppb	7
SnO ₂ /RGO	SWASV	Cd ²⁺ (0.3–1.2); Pb ²⁺ (0.3–1.2); Cu ²⁺ (0.2–0.6); Hg ²⁺ (0.4–1.2) μM	Cd ²⁺ (0.102); Pb ²⁺ (0.184); Cu ²⁺ (0.227); Hg ²⁺ (0.279) nM	39
Pd/PDMS	SWCSV	Mn ²⁺ (0.455–10.9) μM	334 nM	49
MWCNTs/NA/Bi/SPE	DPSV	Zn ²⁺ (0.5–100); Cd ²⁺ (0.5–80); Pb ²⁺ (0.05–100) mg L ⁻¹	---	50
Bi/Carbon	SWASV	Cd ²⁺ (5); Pb ²⁺ (10) ppb	---	51
Bi/Nafion/PPy/MES/GCE	SWASV	Cd ²⁺ (0.05–35); Pb ²⁺ (0.1–25) g L ⁻¹	Cd ²⁺ (0.03); Pb ²⁺ (0.04) g L ⁻¹	52
PPy-CNSs	SWASV	Pb ²⁺ (1–7); Hg ²⁺ (5–35) nM	Pb ²⁺ (0.041); Hg ²⁺ (0.021) nM	53
BONPs-IL-CPE	SWASV	Cd ²⁺ (3.0–30); Pb ²⁺ (3.0–30) μg L ⁻¹	Cd ²⁺ (0.15); Pb ²⁺ (0.21) μg L ⁻¹	61
Pd _{1.5} /PAC-900	DPV	Cd ²⁺ (0.5–5.5); Pb ²⁺ (0.5–8.9); Cu ²⁺ (0.5–5.0); Hg ²⁺ (0.24–7.5) μM	Cd ²⁺ (41); Pb ²⁺ (50); Cu ²⁺ (66); Hg ²⁺ (54) nM	This work

* SNAC: spherical carbon nanoparticle decorated activated carbon; GCE: glassy carbon electrode; HAP: hydroxyapatite; Ag NP-ZSM-5: silver nanoparticles decorated on ZSM-5 zeolite; RGO: reduced graphene oxide; PDMS: poly-dimethylsiloxane; MWCNTs/NA: multi-walled carbon nanotubes nanoarrays; SPE: screen printed electrode; PPy: polypyrrole; CNSs: carbonaceous nanospheres; BONPs-IL-CPE: bismuth oxide nanoparticles and ionic liquid modified carbon paste electrode; SWSV: square wave stripping voltammetry; SWASV: square wave anodic stripping voltammetry.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acsami.xxxxxxx..

Assorted structural parameters and experimental results obtained from XRD, N₂ physisorption, DFT pore-size distribution, DTA, EDX, FT-IR, UV-vis, and DPV studies (PDF)

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Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

The authors are grateful for the financial support (NSC 104-2113-M-001-019 to S.B.L.; NSC 101-2113-M-027-001-MY3 to S.M.C.) from the Ministry of Science and Technology (MOST), Taiwan.

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